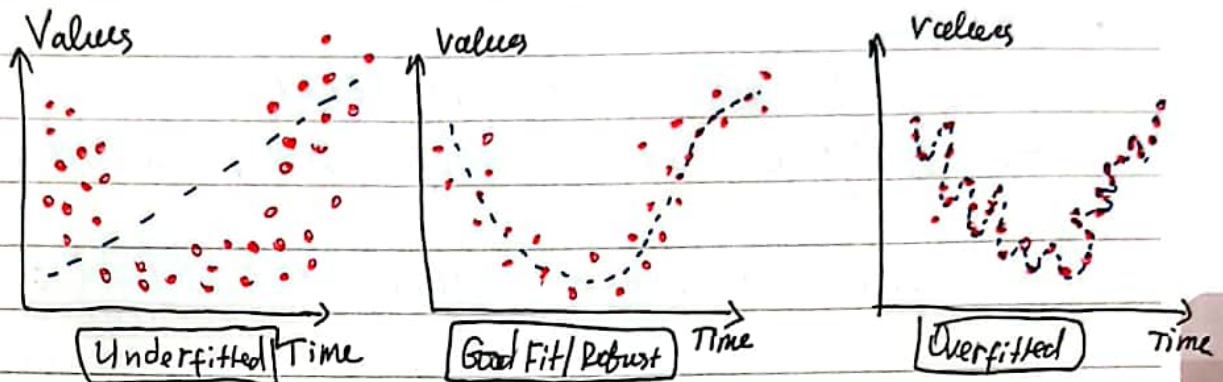


5-dars:



Overfitting va Underfitting ni aniqlash.

- Model yaradilgiz va mse , $rmse$ juda past.
- Lekin testingda mantiqiz javob beryapti yoki yuqori error.
- Bu hda **overfitting** muammosini o'rganishni va uni hal qilishni talab etadi.

Overfitting — — — Yodlab olish
Generalization — — — Umumlashtirish (Tushanish)

Memorization

Short-term

Stored in working memory

Unconnected to real learn

Information is available for quick recall

Limited to words and basic sentences

Limited interpretation only

Does not lead to insights, creativity, or new ways of thinking about the information

Understanding

Long-term or permanent

Stored in long-term memory

Foundational to ~~learn~~ ^{real} learning

Information is available for real world application

Extends to processes, systems and concepts (not just basic vocab)

Deeper, broader, figurative, interpret ^{from} leads to making connections, having insights, synthesizing the information.

- Model to'g'ri ketma-ketliklarni kam xato ketma-ketliklarni o'rganadi (yodlab oladi)
- Qanday bilamiz?
 - ↳ Juda past training error
 - ↳ Yuqori error new/ unseen data da

Overfitting bo'lishini asosiy sabablari:

I. Juda chuqur va ko'p ~~materiallardan~~ ^{parametrlardan} foydalanish.

II. Kam featurelar va kam data

III. Data ko'p lekin tozalanmagan data.

Underfitting nima?

- Featurelar oz bo'lganda va model yetarli darajada datalarni o'rgana olmaganda.
- Belgilari:
 - ↳ Past accuracy, Yuqori training error
 - ↳ Yuqori testing error.

Overfitting da:

I. Algorithm kuchini kamaytirish

II. Ko'prog training data

III. Feature selection / dimensionality reduction

IV. Regularization ($\alpha 1 / \alpha 2$)

V. Dropout in neural networks.

Underfitting da:

I. Increase model complexity (add layers, non-linear models).

II. Use more relevant features.

III. Reduce regularization.

IV. Train longer / optimize hyperparameters.

★ Training ham Test ham past score bōlsa, ⇒
Underfitting

★ Training baland ammo Test past score
bōlsa va orasi juda katta bōlsa ⇒ Overfitting

★ Itkolasli ham normal holat bōlsa ⇒
Good Fit / Robustness holati bōladi.

"Parameters are learned from the data";

"Hyperparameters control how the model learns"

No. Explainable vs Non Explainable

Qilinayotgan ish/ni, olinayotgan natija/ni va chiqarilayotgan modelni aniqligini va predictionni qanday olayotganimizni tushuntirib turadi. Ya'ni Explainable bo'ladi.

Endi, bunga qarama-qarshi ma'nodagi bitta tushuncha bor, dsha sanab o'tkan ishlarimizni qanday qilinayotkanini bilmaymiz biz.